Sanchit Das

Curriculum Vitae

Senior Research Scholar
Department of Mathematics and Computing
Indian Institute of Technology (ISM) Dhanbad
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Research Interests

Core Areas	Applied Mathematics, Continuum Mechanics, Elastic Wave Propagation
Multiphysics Media	Piezoelectric, Poroelastic and Piezo-Magneto-Electric Materials; Layered and Functionally Graded Structures
Modelling	Nonlocal Continuum Models, Viscoelasticity, Imperfect Interfaces, Wave Dispersion and Attenuation
Methods	Analytical, Semi-Analytical and Numerical Methods

Education

2022–Present **Integrated PG–Ph.D.**, *Indian Institute of Technology (ISM) Dhanbad*, Dhanbad, India
Ph.D. in Applied Mathematics – expected completion: June 2027.
M.Sc. in Mathematics and Computing – CGPA: **7.71/10**.

2018–2021 **Bachelor of Science**, *Lucknow Christian Degree College*, University of Lucknow, India
Mathematics, Computer Science and Statistics.

Publications and Preprints

August, 2026 Abhishek Kumar Singh, **Sanchit Das***, Sandeep Sharma. **A Unified Analysis of Love and Bleustein–Gulyaev Waves in Piezo-Viscoelastic Layered Structures with Graded Buffer Layer and Imperfect Interfaces.** Accepted for publication in *The European Physical Journal Plus* (Q2; Impact Factor: 2.9), August 2026, DOI: 10.1140/epjp/s13360-026-08224-0.

May 2026 Abhishek Kumar Singh and **Sanchit Das***. **Impact of buffer layer on piezo-magneto-electric fibre-reinforced composite structure on electro-magneto-mechanical coupling and mass-loading sensitivity analysing Love wave.** *Smart Structures and Systems*, 37(5), 451–479 (Q2). DOI: 10.12989/sss.2026.37.5.451.

April 2026 Abhishek Kumar Singh and **Sanchit Das***. **Nonlocal Love Wave Propagation in Piezoelectric–Poroelastic Layered Media with Spring–Membrane Interfaces and Diverse Surface Exposures.** *International Journal of Numerical Methods for Heat & Fluid Flow*, 36(5), 2198–2225 (Q1; Impact Factor: 5.3). DOI: 10.1108/HFF-12-2025-0971.

April 2026 Abhishek Kumar Singh and **Sanchit Das***. **Non-locality effects on the propagation of Bleustein–Gulyaev wave in a piezo-poroelastic layered structure analyzing spring and membrane models.** *Physica Scripta*, 101, 155212 (Q2; Impact Factor: 2.6). DOI: 10.1088/1402-4896/ae5aee.

*Corresponding author.

Research Experience

June–July 2023 **Summer Research Fellow**, Indian Academy of Sciences (IAS), Indian National Science Academy (INSA), and The National Academy of Sciences, India (NASI).
Project: "Development of Laplacian eigenfunctions through integral operator method."
Worked under the supervision of Dr. S. Janakiraman, Centre for Development of Advanced Computing (C-DAC), Bengaluru.

Teaching Experience

2024–2026 **Numerical Methods Practical**, B.Tech (8th Semester), IIT (ISM) Dhanbad. Teaching Assistant in Monsoon 2024, 2025 and 2026.

2024–2026 **Advanced Numerical Methods Practical**, M.Sc. (1st Semester), IIT (ISM) Dhanbad. Teaching Assistant in Monsoon 2024, 2025 and 2026.

Monsoon 2024 **Computer Organisation and Architecture Practical**, B.Tech (1st Semester), IIT (ISM) Dhanbad.

Winter 2025 **Data Structure Lab**, B.Tech (2nd Semester), IIT (ISM) Dhanbad.

Winter 2025 **Database Management System Tutorial**, M.Sc. (2nd Semester), IIT (ISM) Dhanbad.

Winter 2026 **Database Management System Practical**, B.Tech (4th Semester), IIT (ISM) Dhanbad.

Conferences and Presentations

June 2026 **International Conference on Composite Structures (ICCS29)**, Faculty of Architecture, University of Cagliari, Italy.
Presented: "Liouville–Green Analysis of Love Wave Dynamics at Spring–Membrane Interfaces in Generalised Linearly Graded Piezo-Viscoelastic Structures with Flexoelectric and Semiconductor Coupling."

December 2024 **International Conference on Mechanics of Advanced Materials and Structures**, RIT Bangalore, Texas A&M University, USA, and Politecnico di Torino, Italy.
Presented: "Analysis of Bleustein–Gulyaev and Love Waves in a Composite Structure composed of Magneto-electro-mechanical transversely isotropic materials with a sandwiched buffer layer."

June 2024 **National Conference on Mathematical Modelling and Simulations (MAMS 2024)**, IIT (ISM) Dhanbad.
Presented: "Propagation Characteristics of Love wave in a functionally graded Piezo-electro-magnetic layered composite structure: Power Series Technique."

October 2023 **International Conference on Applied Mathematics and Mechanics**, IIT Indore. Poster presentation.

Awards and Recognitions

2023 **Best Research Paper Award – Poster Category**, International Conference on Applied Mathematics and Mechanics, IIT Indore.

2018 **Best Speaker Award**, Techno-Quest'18, Computer Society of India, Lucknow Chapter, for a presentation on *BlueEyes Technology*.

Technical Skills

Programming Python, C, C++, Java, MATLAB, Wolfram.

Databases SQL, MySQL, MariaDB.

Workshops, Courses and Professional Development

January 2026 Participated in the **Centenary National Workshop on "Bridging Mathematics and Earth Sciences: Tools for Exploration, Modeling, and Analysis – 2026"**, organized by the Department of Applied Geophysics, IIT (ISM) Dhanbad, 17–18 January 2026.

September 2025 Completed the one-week GIAN course **"Introduction to Hydrodynamic Stability"**, IIT (ISM) Dhanbad. Course Coordinator: Prof. Sergey A. Suslov, Swinburne University of Technology, Australia.

February 2024 Participated in the five-day workshop **"Advances in Fluid-Solid Interactions using AI/ML with Special Reference to Numerical Weather Prediction and Smart Farming using IoT (AFSI-2024)"**, IIT (ISM) Dhanbad.

Academic Service and Volunteering

June 2024 **Executive Head**, Delegates and Faculty Guest Management Committee, MAS Conference 2024, IIT (ISM) Dhanbad.

2023–2024 **Volunteer, Media and Branding**, Karma Jyoti, IIT (ISM) Dhanbad.

2022–2023 **Class Representative**, M.Sc. (Mathematics and Computing), Batch of 2024, IIT (ISM) Dhanbad.

Referees

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